

D2 Requirements Report

Kitchen Display System (KDS)

CSE323 - Software Engineering

Project: Ctrl-Alt-Eat

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1. Introduction

The Kitchen Display System is the operational part of the Ctrl-Alt-Eat restaurant application. Its main job is to move customer orders from the menu/cart flow into a live kitchen dashboard where chefs can prepare orders in arrival order and update their status. The system reduces the need for handwritten tickets or verbal communication between the front side of the restaurant and the kitchen.

The current implementation uses a React frontend and a Django REST Framework backend. Customers can sign up, browse menu items, select add-ons, place an order for a table, and track their order. Chefs can view active orders in FIFO order and update them from in_progress to ready, then to delivered. WebSocket notifications are used for ready-order alerts, with frontend filtering so customers only see notifications for their own orders.

The scope of this report is limited to Phase 1: requirement discovery and traceability. It does not evaluate the full design or validation deliverables except where tests are useful as traceability evidence.

2. Actor Classification

2.1 Primary Actors

Primary actors directly use the system to complete their own goals.

Actor	Description	Responsibilities	Why This Classification Fits
Customer	A restaurant customer using the web interface to order food.	Create/login to an account, browse menu categories, choose add-ons, manage cart, submit orders, track status, view history.	The customer starts the main business flow by creating an order. Without this actor, the kitchen board has no incoming work.
Chef / Kitchen Staff	A kitchen user responsible for preparing and completing orders.	Monitor live orders, read item details and notes, mark orders as ready, mark ready orders as delivered.	This actor receives the main output of the ordering flow and uses the dashboard to perform kitchen work.

2.2 Supporting Actors

Supporting actors provide services that help the primary actors complete their goals.

Actor	Description	Responsibilities	Why This Classification Fits
Authentication Service	The Django token authentication layer used by the API and frontend.	Validate login credentials, issue tokens, support logout and /auth/me/ session checks.	It supports customer and chef access but does not use the KDS for its own business goal.
Database	The persistence layer behind Django models.	Store users, roles, menu items, add-ons, orders, order items, and order status.	It supports all flows by keeping system state consistent across requests.
WebSocket Channel Layer	Django Channels in-memory channel layer.	Broadcast order-ready events to connected clients.	It supports real-time feedback, but it is not a human user of the system.
React Frontend	The browser application used by customers and chefs.	Route users by role, collect input, display menus/orders, call APIs, show errors and notifications.	It mediates interaction between users and backend services.

2.3 Offstage Actors

Offstage actors care about the outcome but are not directly part of the main runtime workflow.

Actor	Description	Responsibilities / Interest	Why This Classification Fits
Restaurant Manager / Admin	A manager who needs the restaurant to process orders correctly.	Maintain menu data, observe order flow, rely on accurate kitchen operation.	The implementation includes admin/staff roles and Django admin support, but the manager is not the main dashboard user.
Cashier / Payment Role	A possible restaurant role around payments and checkout.	Care about payment method selection and completed orders.	A cashier role exists in the model and payment method UI exists, but there is no complete cashier workflow in this subsystem.
IT / System Maintainer	Person responsible for deployment and backend configuration.	Configure API URL, CORS, database, server process, and WebSocket runtime.	They support operation but do not participate in order preparation.
Course Evaluator	The grader reviewing whether requirements, traceability, and edge cases are properly documented.	Inspect submitted reports and implementation evidence.	This actor influences documentation quality but is not part of the restaurant workflow.

3. Requirements

The following requirements are extracted only from the implemented project. They avoid features that are not present in the code.

3.1 Functional Requirements

ID	Requirement	Priority	Related Actor(s)	Implementation Evidence
FR-01	The system shall allow users to sign up and log in using email/password credentials.	High	Customer, Chef	SignupSerializer, LoginSerializer, SignupView, LoginView, AuthContext
FR-02	The system shall classify a new user as chef when the email ends with @ejust.edu.eg; otherwise the user is a customer.	High	Customer, Chef	SignupSerializer.create, Login.jsx detected role preview, Auth.test.jsx
FR-03	The frontend shall protect routes based on user role.	High	Customer, Chef	ProtectedRoute in App.jsx, customer and chef route definitions
FR-04	The system shall expose menu items grouped by category, including available add-ons.	High	Customer	MenuItemViewSet.list, MenuItem, AddOn, CreateOrder.jsx
FR-05	Customers shall be able to add menu items to a cart with quantities and selected add-ons.	High	Customer	CartContext.jsx, CreateOrder.jsx, Cart.test.jsx
FR-06	Customers shall be able to place an order containing table number and cart items.	High	Customer, Database	OrderCreateSerializer, OrderViewSet.create, Cart.jsx, CheckoutPage.jsx
FR-07	The backend shall store each order with its creator, table number, items, quantities, add-ons, notes, status, priority, and creation time.	High	Customer, Chef, Database	Order, OrderItem, OrderSerializer
FR-08	Customers shall only see their own orders through the orders API and order history page.	High	Customer	OrderViewSet.get_queryset, OrderHistory.jsx, integration test
FR-09	Chefs shall see active kitchen orders in FIFO order.	High	Chef	DashboardViewSet.get_queryset, Dashboard.jsx, KitchenBoard.jsx
FR-10	Chefs shall update order status from in_progress to ready, then from ready to delivered.	High	Chef, Customer	OrderViewSet.partial_update, StatusButtons.jsx, integration tests
FR-11	Customers shall receive a ready-order notification for their own order.	Medium	Customer, WebSocket Channel	OrderConsumer, useOrderSocket, App.jsx customer-id filtering, Toast.jsx
FR-12	The system shall support logout and token/session cleanup on the frontend.	Medium	Customer, Chef	LogoutView, AuthContext.logout, Navbar.jsx

ID	Requirement	Priority	Related Actor(s)	Implementation Evidence
FR-13	The checkout UI shall allow selecting a demo payment method before placing an order, without processing real payment.	Low	Customer, Cashier	PaymentOptions.jsx, Cart.jsx, CheckoutPage.jsx

3.2 Non-Functional Requirements

These requirements describe quality constraints and operating conditions that the implementation already addresses to some level.

ID	Non-Functional Requirement	Quality Attribute	Priority	Implementation Evidence
NFR-01	The system shall enforce role-based access control so customers cannot access kitchen-only operations.	Security	High	ProtectedRoute in App.jsx, OrderViewSet.get_queryset, dashboard/status permission checks
NFR-02	The system shall protect data integrity by rejecting invalid table numbers, invalid quantities, negative prices, and overly long notes.	Data Integrity	High	Django validators in models.py, serializer validation tests
NFR-03	The customer notification flow shall recover from temporary WebSocket disconnection where possible.	Reliability	Medium	useOrderSocket reconnect logic, OrderStatusPoller fallback polling
NFR-04	The interface shall provide clear feedback for empty cart, empty menu category, empty kitchen queue, loading, and failed request states.	Usability	Medium	Cart.jsx, CreateOrder.jsx, Dashboard.jsx, KitchenBoard.jsx
NFR-05	The kitchen dashboard shall keep active orders understandable during busy periods by preserving FIFO ordering and showing simple workload indicators.	Operational Performance	Medium	DashboardViewSet.get_queryset, Dashboard.jsx active counts, KitchenBoard.jsx wait-time display

4. Traceability Heatmap / Matrix

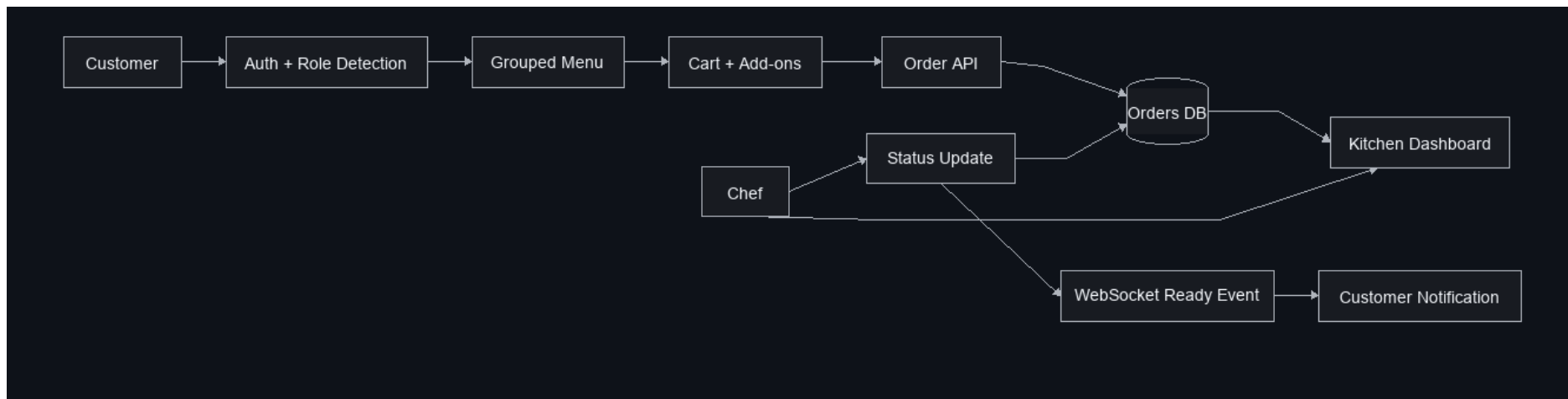
Legend: GREEN / H = strong direct evidence, YELLOW / M = partial evidence, RED / L = weak/supporting evidence, - = not applicable.

Heat Level	Meaning	Interpretation
GREEN / H	Strong coverage	Requirement is directly supported by UI, backend/data evidence, and tests or strong implementation proof.
YELLOW / M	Medium coverage	Requirement is implemented, but evidence is partial or would benefit from more tests.
RED / L	Weak coverage	Requirement has limited evidence, usually because it is UI-only or outside the core backend flow.

Req ID	Feature / Behavior	UI Screen or Component	API / Backend	Database Entities	Tests / Evidence	Coverage
FR-01	Signup/login	Login.jsx	/api/auth/signup/, /api/auth/login/	User, Token	test_signup_api, test_login_api, Auth.test.jsx	GREEN / H
FR-02	Role detection	Login.jsx role preview	SignupSerializer.create	User.role	test_signup_serializer_customer, test_signup_serializer_chef, Auth.test.jsx	GREEN / H
FR-03	Role routing	App.jsx, Navbar.jsx	Authenticated API calls	User.role	Frontend context tests indirectly	YELLOW / M
FR-04	Menu browsing	CreateOrder.jsx	GET /api/menu-items/	MenuItem, AddOn	test_menu_items_api	GREEN / H

Req ID	Feature / Behavior	UI Screen or Component	API / Backend	Database Entities	Tests / Evidence	Coverage
FR-05	Cart management	CreateOrder.jsx, Cart.jsx	-	Browser state only	Cart.test.jsx	GREEN / H
FR-06	Order submission	Cart.jsx, CheckoutPage.jsx	POST /api/orders/	Order, OrderItem	test_create_order_authenticated, serializer tests	GREEN / H
FR-07	Order persistence	Kitchen/order screens	OrderSerializer	Order, OrderItem, MenuItem, AddOn, User	test_order_model, test_order_item_model	GREEN / H
FR-08	Customer order isolation	OrderHistory.jsx	OrderViewSet.get_queryset	Order.created_by	test_customer_cannot_see_others_orders	GREEN / H
FR-09	FIFO kitchen dashboard	Dashboard.jsx, KitchenBoard.jsx	GET /api/dashboard/	Order.created_at, Order.order_status	Integration tests support endpoint access	YELLOW / M
FR-10	Status workflow	StatusButtons.jsx	PATCH /api/orders/{id}/	Order.order_status	test_chef_can_update_status, access-control tests	GREEN / H
FR-11	Ready notification	Toast.jsx, ReadyNotification.jsx	ws/orders/, group broadcast	Order.created_by_id in event	test_order_ready_broadcast	YELLOW / M
FR-12	Logout/session cleanup	Navbar.jsx	/api/auth/logout/, /api/auth/me/	Token, User	Auth context tests partly cover state changes	YELLOW / M
FR-13	Demo payment choice	PaymentOptions.jsx	Not persisted	-	UI component evidence only	RED / L
NFR-01	Role-based access control	App.jsx protected routes	Dashboard/status permission checks	User.role, Order.created_by	Customer access restriction tests	GREEN / H
NFR-02	Data integrity boundaries	Forms and API errors	Serializers/models	Order, OrderItem, MenuItem, AddOn	Model and serializer boundary tests	GREEN / H
NFR-03	Notification recovery	OrderTrackingPage.jsx	WebSocket + polling endpoints	Order.order_status	WebSocket test, component evidence	YELLOW / M
NFR-04	Empty/loading/error feedback	Cart/menu/kitchen screens	API error handling in services/components	-	Manual component evidence	YELLOW / M
NFR-05	Busy-period readability	Dashboard.jsx, KitchenBoard.jsx	FIFO dashboard query	Order.created_at, Order.order_status	Partial endpoint evidence	YELLOW / M

4.1 Requirement-to-Implementation Diagram



4.2 Orphan Check

Item Checked	Result
Requirements without implementation evidence	None in this D2 report. Low evidence is clearly marked instead of hidden.
Implemented features without requirement mapping	Payment method UI is mapped as FR-13 and marked low because it is UI-only.
Features with missing tests	Route protection, empty UI states, FIFO ordering, and demo payment selection would benefit from more frontend/e2e tests.
Requirements that are outside current implementation	Real payment processing, cashier dashboard, printer integration, and production-grade WebSocket auth are intentionally not claimed.

5. Persona Discovery

Persona 1: Omar, Busy Lunch Customer

Omar is a student ordering food between classes. He wants to select a meal quickly, avoid waiting at the counter, and know when the order is ready. He is comfortable using a web app but will not read long instructions. His main pain point is uncertainty: he dislikes not knowing whether the kitchen actually received the order.

Persona 2: Sara, Line Chef

Sara works during peak hours and needs a simple board that shows the oldest active orders first. She cares about table number, item details, add-ons, and special notes. Her pain point is missed modifications, especially when several orders arrive at once.

Persona 3: Mina, Restaurant Supervisor

Mina is responsible for keeping service moving. He is not always the person touching the dashboard, but he cares about bottlenecks, stuck orders, wrong role access, and whether the system can recover from common failures.

5.1 Hidden Requirements and Edge Cases

ID	Edge Case / Hidden Requirement	Scenario	Risk	Current Handling	Limitation
EC-01	Unauthorized order status update	A customer tries to call PATCH /api/orders/{id}/ and mark their own order ready.	Fake ready states and broken kitchen trust.	Backend blocks status updates unless the user is chef/admin/staff; frontend also restricts chef dashboard routes.	More negative frontend tests would improve evidence.
EC-02	Customer opens kitchen dashboard API	A logged-in customer manually requests /api/dashboard/.	Customer can see other tables or kitchen workload.	Backend returns 403 for non-kitchen staff.	This should remain covered in integration tests after future changes.
EC-03	Lost WebSocket connection	A customer's browser disconnects before the order becomes ready.	Customer may miss the ready alert.	useOrderSocket reconnects after 3 seconds, and the tracking page polls order status.	WebSocket auth is still simple; production would use authenticated per-user groups.
EC-04	Notification broadcast to unrelated customers	Multiple customers are connected when one order becomes ready.	Wrong customer receives a ready message.	The frontend checks event.customer_id against the logged-in user id before showing a toast.	The event is still broadcast to a shared group, but it does not include item details.
EC-05	Simultaneous status changes	Two kitchen users click status buttons on the same order almost at the same time.	Order could skip states or end in a confusing status.	Backend status-transition rules only allow valid next states.	There is no optimistic locking/version field, so the last valid request still wins.
EC-06	Duplicate order submission	A customer double-clicks or retries during slow network.	Same cart may be submitted twice.	UI disables the submit button while a request is in progress.	There is no backend idempotency key, so refresh/retry duplicates are still possible.

ID	Edge Case / Hidden Requirement	Scenario	Risk	Current Handling	Limitation
EC-07	Invalid table number or quantity	User submits table 0, table 101, quantity 0, or quantity 51.	Bad kitchen tickets or impossible values in reports.	Django validators and serializer tests reject invalid boundaries.	Frontend validation is lighter than backend validation, so API remains the main guard.
EC-08	Empty queue / empty cart / empty category	The kitchen has no active orders, the cart is empty, or a category has no items.	User may think the app is broken.	Components show specific empty states and navigation actions.	Empty states are manually implemented but not fully covered by e2e tests.
EC-09	Kitchen overload	Many orders arrive during rush hour.	Chefs may lose the preparation order or miss long-waiting tickets.	Dashboard sorts FIFO and shows active counts plus waiting time.	No automatic escalation, SLA timer, or priority override UI exists yet.
EC-10	Weak role proof through email domain	A user signs up with an @ejust.edu.eg address and becomes chef.	If email ownership is not verified, role assignment can be abused.	The implementation uses email-domain detection as required by the demo flow.	A real deployment should require admin approval or verified institutional login.

6. Requirement Coverage Evaluation

For Phase 1, the project is now close to the rubric's Excellent band:

Rubric Criterion	Evaluation
Actor Classification	All three actor types are identified with rationale. Primary, supporting, and offstage actors are separated clearly.
Traceability Heatmap	Every requirement in this report maps to implementation evidence. The matrix also marks weak/partial areas instead of overstating them.
Persona Discovery	More than five edge cases are derived from realistic customer, chef, and supervisor personas. Each case includes scenario, risk, current handling, and limitation.

The main remaining weakness is not the D2 report itself, but test depth around frontend route protection, empty-state UI, and production-grade WebSocket scoping. These are documented honestly and do not require inventing features that are absent from the project.

7. Conclusion

The Kitchen Display System implements a practical vertical slice for restaurant order handling: customers submit orders, the kitchen receives active tickets, chefs update status, and customers receive readiness feedback. The requirements can be traced through React screens, Django APIs, database models, and available tests. A few production features are intentionally outside the current scope, but the core Phase 1 requirements are now documented with clear actor classification, traceability, and persona-driven edge cases.